

PROBLEM NO: 1

NAME OF THE PROBLEM: Write a program using Python for addition and multiplication of two numbers.

SOURCE CODE:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

addition = a + b
multiplication = a * b

print("Addition =", addition)
print("Multiplication =", multiplication)
```

OUTPUT:

```
Enter first number: 10
Enter second number: 20
Addition = 30
Multiplication = 200
```

PROBLEM NO: 2

NAME OF THE PROBLEM: Write a program using Python for finding the sum of all numbers in a given list.

SOURCE CODE:

```
numbers = [10, 20, 30, 40, 50]
total = 0

for n in numbers:
    total = total + n

print("List:", numbers)
print("Sum of all numbers =", total)
```

OUTPUT:

List: [10, 20, 30, 40, 50] Sum of all numbers = 150

PROBLEM NO: 3

NAME OF THE PROBLEM: Write a program using Python for comparing character and string.

SOURCE CODE:

```
ch = input("Enter a character: ")
text = input("Enter a string: ")

if ch == text:
    print("Character and string are equal.")
else:
    print("Character and string are not equal.")

if ch in text:
    print("The character is present in the string.")
else:
    print("The character is not present in the string.")
```

OUTPUT:

```
Enter a character: a
Enter a string: apple
Character and string are not equal.
The character is present in the string.
```

PROBLEM NO: 4

NAME OF THE PROBLEM: Write a program using Python to count number of elements in a list.

SOURCE CODE:

```
items = [5, 10, 15, 20, 25, 30]
count = 0

for item in items:
    count = count + 1

print("List:", items)
print("Number of elements =", count)
```

OUTPUT:

```
List: [5, 10, 15, 20, 25, 30]
Number of elements = 6
```

PROBLEM NO: 5

PROBLEM NAME: Write a program using Python whether an element is a member of list.

SOURCE CODE:

```
items = [5, 10, 15, 20, 25, 30]
count = 0
for item in items:
    count = count + 1
print("List:", items)
print("Number of elements =", count)
```

OUTPUT:

```
Enter element to search: 9
9 is a member of the list.
```

PROBLEM NO: 6

PROBLEM NAME: Write a program using Python to reverse a list.

SOURCE CODE:

```
numbers = [1, 2, 3, 4, 5]
reversed_list = []

for item in numbers:
    reversed_list.insert(0, item)

print("Original list:", numbers)
print("Reversed list:", reversed_list)
```

OUTPUT:

```
Original list: [1, 2, 3, 4, 5]
Reversed list: [5, 4, 3, 2, 1]
```

PROBLEM NO: 7

PROBLEM NAME: Write a program using Python to find out Union and Intersection of two lists.

SOURCE CODE:

```
list1 = [1, 2, 3, 4, 5]
list2 = [4, 5, 6, 7, 8]

union = []
intersection = []

for item in list1 + list2:
    if item not in union:
        union.append(item)

for item in list1:
    if item in list2:
        intersection.append(item)

print("First list:", list1)
print("Second list:", list2)
print("Union:", union)
print("Intersection:", intersection)
```

OUTPUT:

```
First list: [1, 2, 3, 4, 5]
Second list: [4, 5, 6, 7, 8]
Union: [1, 2, 3, 4, 5, 6, 7, 8]
Intersection: [4, 5]
```

PROBLEM NO: 8

PROBLEM NAME: Write a program using Python to determine the Greatest Common Divisor of two positive integer numbers.

SOURCE CODE:

```
a = int(input("Enter first positive integer: "))
b = int(input("Enter second positive integer: "))

x = a
y = b

while y != 0:
    r = x % y
    x = y
    y = r
print("GCD of", a, "and", b, "=", x)
```

OUTPUT:

```
Enter first positive integer: 48
Enter second positive integer: 18
GCD of 48 and 18 6
```